

NGSO Satellites Beam Hopping Strategy Based on Load Balancing and Interference Avoidance for Coexistence With GSO Systems

Zhiyuan Lin¹, Graduate Student Member, IEEE, Zuyao Ni¹, Member, IEEE, Linling Kuang¹, Member, IEEE, Chunxiao Jiang¹, Senior Member, IEEE, and Zhen Huang, Member, IEEE

Abstract—Non-geostationary orbit (NGSO) satellite communication system plays an important role in achieving seamless global coverage and addressing the digital divide. The uneven distribution of terrestrial users and the high mobility of NGSO satellites put forward higher requirements for resource management. In this letter, we propose a multi-satellite beam hopping scheme based on load balancing and interference avoidance, where the complicated multi-satellite joint beam hopping problem is decomposed into two sub-problems, namely the satellite-cell assignment problem and joint multi-satellite beam hopping pattern design problem. On the one hand, we rationally assign cells to different satellites for balancing the traffic load among satellites and reducing the interference to geostationary earth orbit (GSO) ground stations. On the other hand, an iterative multi-satellite beam hopping pattern design scheme based on minimizing co-channel interference criterion is adopted to achieve on-demand service. Simulation results indicate that the performance of load balancing and interference avoidance are improved compared to the maximum elevation angle access strategy. Besides, the proposed scheme outperforms other benchmarks in terms of the traffic satisfaction rate, and the optimality gap is only 0.1%.

Index Terms—Beam hopping, load balancing, interference avoidance, NGSO satellite communications, resource allocation.

I. INTRODUCTION

RECENTLY, a lot of non-geostationary orbit (NGSO) constellations (e.g., OneWeb, Starlink and Telesat) have been developed rapidly owing to the ability to provide seamless coverage and broadband service [1]. Due to the uneven geographic distribution of users and high mobility of NGSO satellites, the traffic demands reveal temporal and spatial variability. Under this circumstance, how to match the limited satellite resources to the non-uniform demands is a significant challenge for NGSO satellite systems [2].

To address this issue, beam hopping (BH) is envisioned as a flexible means to achieve on-demand service, and there are already three super frame structures (e.g., Format 5-7)

defined in DVB-S2X standard to support BH technique [3]. Nonetheless, how to design effective BH patterns to match non-uniform traffic is challenging. In [4], an iterative BH scheme based on maximizing signal to interference plus noise (SINR) was proposed. In addition, [5] proposed a greedy strategy for beam scheduling with various mission. Besides, some scholars introduced deep learning (DL) into BH. [6] proposed a combined DL-and-optimization method to achieve a fast BH scheduling. In order to perform traffic-driven BH, a multi-agent deep reinforcement learning framework was proposed in [7] to simultaneously determine the irradiation direction and bandwidth of beams. Furthermore, [8] proposed a greedy beam scheduling scheme for NR-based LEO satellite constellation.

However, above works focus on BH for a single satellite or do not consider the coexistence with GSO system. Motivated by this point, in this letter, we study multi-satellite beam hopping strategy for NGSO satellite system to realize on-demand service and avoid interference to GSO ground stations. The key contributions of this letter can be summarized as follows:

- An efficient satellite-cell allocation scheme is proposed to balance the traffic load among satellites and avoid interference to GSO ground stations, which is formulated as a simple mixed integer linear programming.
- Based on the criterion of minimizing inter-beam interference, this letter proposes a low-complexity iterative multi-satellite BH pattern design approach to match uneven traffic demands as much as possible.
- The performance of the proposed method is investigated herein in terms of satellite load, traffic satisfaction rate and interference. Simulation results demonstrate that the proposed technique outperforms other benchmarks.

II. SYSTEM MODEL AND PROBLEM FORMULATION

A. System Model

This letter considers a typical BH scenario in the downlink segment of the forward link, as shown in Fig. 1. Assuming that a target area contains N cells, all of which can be covered by M NGSO satellites. Let $\mathcal{N}$ and $\mathcal{M}$ represent the set of cells and NGSO satellites. Besides, suppose there are L GSO ground stations distributed in different cells. We denote $\mathcal{L}$ as the set of GSO ground stations. Generally, the downlink between NGSO satellites and NGSO users should not interfere with the communication between GSO satellite and GSO ground stations. For convenience, the satellites mentioned later refer to NGSO satellites.

In BH system, each satellite can form no more than K spot-beams simultaneously to sever different cells in a time-division multiplexing manner. Besides, the size of each cell is the

Manuscript received 28 September 2022; accepted 10 October 2022. Date of publication 12 October 2022; date of current version 9 January 2023. This work was supported by the National Key Research and Development Program under Grant 2019YFB1803005. The associate editor coordinating the review of this letter and approving it for publication was Q. Wu. (Corresponding author: Zuyao Ni.)

Zhiyuan Lin is with the Department of Electronic Engineering and the Beijing National Research Center for Information Science and Technology, Tsinghua University, Beijing 100091, China (e-mail: zy-lin19@mails.tsinghua.edu.cn).

Zuyao Ni, Linling Kuang, Chunxiao Jiang, and Zhen Huang are with the Tsinghua Space Center and the Beijing National Research Center for Information Science and Technology, Tsinghua University, Beijing 100091, China (e-mail: nzy@tsinghua.edu.cn; kll@tsinghua.edu.cn; jchx@tsinghua.edu.cn; huangzhen@tsinghua.edu.cn).

Digital Object Identifier 10.1109/LCOMM.2022.3213912

1558-2558 © 2022 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission. See <https://www.ieee.org/publications/rights/index.html> for more information.

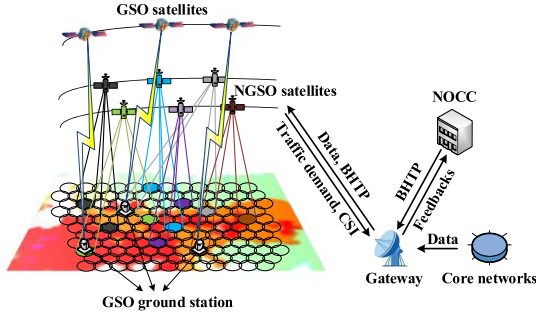


Fig. 1. The forward link of the multi-beam NGSO satellite communication system.

same as that of one spotbeam, and thus each spotbeam can serve only one cell at one timeslot. We denote $\mathcal{K}_m$ as the set of spotbeams for satellite m and we have $|\mathcal{K}_m| \leq K$. The gateway connects the core network and the satellites, and collects the traffic demand and channel condition of each cell through the return link. Due to the non-uniform geographic distribution of users, the traffic demand of each cell is much different. Let D_n denote the traffic demand of cell n . To match these non-uniform demands, the network operation control center (NOCC) needs to decide the beam hopping time plan (BHTP) of satellites. BHTP generally contains W timeslots and the illumination direction of spotbeams at each timeslot is called BH pattern. Let $\mathcal{W}$ represent the set of timeslots. The BHTP of satellite m can be expressed as

$$\mathbf{X}_m = \{\mathbf{x}_m^1, \mathbf{x}_m^2, \dots, \mathbf{x}_m^t, \dots, \mathbf{x}_m^W\}. \quad (1)$$

where $\mathbf{x}_m^t$ denotes the BH pattern of satellite m at timeslot t and can be further represented as $\mathbf{x}_m^t = \{x_{m,1}^t, \dots, x_{m,k}^t, \dots, x_{m,|\mathcal{K}_m|}^t\}$. $x_{m,k}^t \in \mathcal{N}$ indicates which cell is served by the k -th spotbeam of satellite m at timeslot t .

Denote p_{mk}^t as the transmit power for the spotbeam k of satellite m at timeslot t . Assuming full frequency reuse between spotbeams, the SINR of the user k (assuming consistent indexes between users and spotbeams) served by satellite m at timeslot t can be derived as (2), shown at the bottom of the page, where $\sigma^2 = k_B T_{noise} B$ denotes the noise power. k_B , T_{noise} and B represent the Boltzmann constant, the noise temperature of receiver and the bandwidth, respectively. It is noted that the co-frequency interference consists of intra-satellite interference and inter-satellite interference. We denote $|h_{m'k',mk}|^2$ as the channel gain from the k' -th spotbeam of satellite m' to the k -th user of satellite m . This letter use a widely-adopted channel model for BH systems [7], [8], [9], which can be derived as,

$$|h_{m'k',mk}|^2 = \frac{G_{m'k',mk}^{tx} G_{mk,m'}^{rx}}{(4\pi \frac{d_{mk,m'}}{\lambda})^2}, \quad (3)$$

where $G_{m'k',mk}^{tx}$ denotes the transmit antenna gain from the k' -th spotbeam of satellite m' to the k -th user of satellite m , and $G_{mk,m'}^{rx}$ is the receive antenna gain of the user k towards the satellite m' . $d_{mk,m'}$ is the distance between the user k and the satellite m' , and λ refers to the wavelength.

According to (1) and (2), we can obtain the offered capacity for cell n during the whole BHTP as follows,

$$R_n = \frac{1}{W} \sum_{t \in \mathcal{W}} \sum_{m \in \mathcal{M}} \sum_{k \in \mathcal{K}_m} \delta_{x_{m,k}^t=n} B \log_2(1 + \gamma_{mk}^t), \quad (4)$$

where $\delta_{x_{m,k}^t=n}$ is the indicator function, which means that if $x_{m,k}^t = n$, $\delta_{x_{m,k}^t=n} = 1$; otherwise, $\delta_{x_{m,k}^t=n} = 0$.

In terms of the l -th GSO ground station, the interference received from NGSO satellites at timeslot t can be represented as

$$I_l^t = \sum_{m \in \mathcal{M}} \sum_{k \in \mathcal{K}_m} p_{mk}^t |h_{mk,l}|^2, \quad (5)$$

where $|h_{mk,l}|^2$ is the channel gain from the k -th spotbeam of satellite m to the l -th GSO ground station, which has the same form as (3).

B. Optimization Problem Formulation

We apply the ratio of offered capacity against traffic demand (i.e. $\frac{R_n}{D_n}$) to measure the traffic satisfaction rate of each cell. The goal of multi-satellite beam hopping is to maximize the minimum traffic satisfaction rate among the cells while ensuring that each GSO ground station satisfies the interference constraints. Thus, the problem is formulated as,

$$\begin{aligned} \mathcal{P}_0 : \quad & \max_{\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_M} \min_n \frac{R_n}{D_n} \\ & \text{s.t. } C_1 : |\mathcal{K}_m| \leq K, \quad \forall m \in \mathcal{M}, \\ & \quad C_2 : \sum_{k \in \mathcal{K}_m} p_{mk}^t \leq P_{sat}, \quad \forall m \in \mathcal{M}, \quad \forall t \in \mathcal{W}, \\ & \quad C_3 : I_l^t \leq I_{th}, \quad \forall l \in \mathcal{L}, \quad \forall t \in \mathcal{W}. \end{aligned} \quad (6)$$

$\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_M$ are BHTP integer variables to be optimized, which have been defined in (1). C_1 indicates that each satellite can generate up to K spotbeams and C_2 constrains the sum of beam power to be less than the total satellite power P_{sat} . In C_3 , the interference to GSO ground station should meet the maximum interference threshold I_{th} requirements.

It is noted that $\mathcal{P}_0$ is a non-convex integer optimization problem, which is NP-hard. In order to reduce the computational cost, this letter decomposes $\mathcal{P}_0$ into two sub-problems to get a sub-optimal solution, namely the satellite-cell assignment problem and joint multi-satellite BHTP design problem.

$$\gamma_{mk}^t = \frac{p_{mk}^t |h_{mk,mk}|^2}{\sigma^2 + \underbrace{\sum_{k' \in \mathcal{K}_m \setminus \{k\}} p_{mk'}^t |h_{mk',mk}|^2}_{\text{Intra-satellite interference}} + \underbrace{\sum_{\substack{m' \in \mathcal{M} \setminus \{m\} \\ k' \in \mathcal{K}_{m'}}} p_{m'k'}^t |h_{m'k',mk}|^2}_{\text{Inter-satellite interference}}}.$$

(2)

III. PROBLEM DECOMPOSITION AND PROPOSED APPROACH

A. Satellite-Cell Assignment Problem

As shown in Fig. 1, one hotspot area may contain multiple adjacent cells. If these high-demand cells are served by one satellite, this satellite will be severely overloaded. On the other hand, if multiple satellites serve the same hotspot cell simultaneously, it will cause heavy multi-satellite interference. Thus, we can assign hotspot cells to different satellites, which can not only reduce the load pressure of a single satellite, but also avoid serious multi-satellite interference.

When allocating cells to satellites, it is necessary to consider the interference effect from NGSO satellites to GSO ground stations. As shown in Fig. 2, suppose one GSO ground station is located in the n -th cell, and at the same time, the NGSO satellite serves the user of this cell. Denote θ as the angle between the GSO downlink and the NGSO interference link. If $\theta \leq \theta_{th}$ (θ_{th} determined by the 3dB beamwidth of GSO ground station), this NGSO satellite will critically interfere with the GSO ground station. Therefore, the n -th cell is not supposed to be assigned to this NGSO satellite.

In summary, the satellite-cell assignment problem can be described as minimizing traffic load gap between satellites while avoiding interference to GSO ground stations, which has the form as,

$$\begin{aligned} \mathcal{P}_1 : \quad & \min_{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_M} \{ \max_m \{S_m\} - \min_m \{S_m\} \} \\ \text{s.t. } & C_1 : S_m = \mathbf{D}^T \mathbf{u}_m = \sum_{n \in \mathcal{N}} D_n u_{m,n}, \quad \forall m \in \mathcal{M}, \\ & C_2 : u_{m,n} = 0, \quad \forall n \in V_{gs}, \quad \forall m \in G(n), \\ & C_3 : \sum_{m \in \mathcal{M}} u_{m,n} = 1, \quad \forall n \in \mathcal{N}, \\ & C_4 : u_{m,n} \in \{0, 1\}, \quad \forall n \in \mathcal{N}, \quad \forall m \in \mathcal{M}, \end{aligned} \quad (7)$$

where $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_M \in \{0, 1\}^{N \times 1}$ are the binary satellite-cell allocation variables to be optimized. $u_{m,n} = 1$ indicates that cell n is assigned to satellite m . S_m is the total traffic load of satellite m . In C_2 , the cell where the GSO ground station is located cannot be allocated to some satellites due to interference restrictions. V_{gs} represents the set of cells where GSO ground stations are situated, and $G(n)$ is the set of satellites that do not satisfy the interference constraint (i.e., $\theta \leq \theta_{th}$). C_3 confines that each cell can only be allocated to one satellite. Let $\xi \triangleq \max\{S_m\}$ and $\eta \triangleq \min\{S_m\}$. $\mathcal{P}_1$ can be renewed as follow,

$$\begin{aligned} \mathcal{P}_2 : \quad & \min_{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_M} \xi - \eta \\ \text{s.t. } & C_1 : S_m = \mathbf{D}^T \mathbf{u}_m = \sum_{n \in \mathcal{N}} D_n u_{m,n}, \quad \forall m \in \mathcal{M}, \\ & C_2 : u_{m,n} = 0, \quad \forall n \in V_{gs}, \quad \forall m \in G(n), \\ & C_3 : \sum_{m \in \mathcal{M}} u_{m,n} = 1, \quad \forall n \in \mathcal{N}, \\ & C_4 : u_{m,n} \in \{0, 1\}, \quad \forall n \in \mathcal{N}, \quad \forall m \in \mathcal{M}, \\ & C_5 : S_m \leq \xi, \quad \forall m \in \mathcal{M}, \\ & C_6 : \eta \leq S_m, \quad \forall m \in \mathcal{M}. \end{aligned} \quad (8)$$

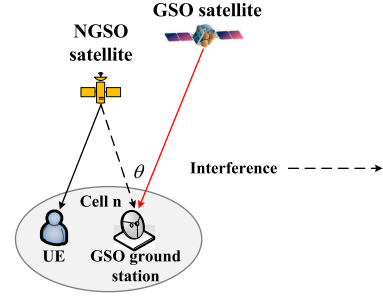


Fig. 2. An illustration of interference from NGSO satellite to GSO ground station.

$\mathcal{P}_2$ is a mixed integer linear programming (MILP), which can be efficiently solved by the branch and bound method in CVX toolbox [10].

Algorithm 1 JIMS-BH

Input: The satellite-cell allocation result $\mathcal{N}_m$.
Initialize: $\forall m \in \mathcal{M}, \mathbf{X}_m = \emptyset$.
1: **repeat**
2: $\forall m \in \mathcal{M}, \mathbf{X}_{m,temp} = \mathbf{X}_m$.
3: Calculate the offered capacity R_n according to (4).
4: Find the cell $\tilde{n}_m$ with the minimum traffic satisfaction rate for each satellite: $\forall m \in \mathcal{M}, \tilde{n}_m = \min_{n \in \mathcal{N}_m} \frac{R_{n,m}}{D_{n,m}}$.
5: **for** $m = 1$ to M **do**
6: **for** $t = 1$ to W **do**
7: **if** $|\mathbf{x}_m^t| < K$ and $\tilde{n}_m \notin \mathbf{x}_m^t$ **then**
8: Assuming that $\tilde{n}_m$ is served by satellite m at timeslot t and the power of satellite is evenly distributed across spotbeams, calculate the interference I_t received by cell $\tilde{n}_m$ according to (2).
9: **else**
10: $I_t = +\infty$.
11: **end if**
12: **end for**
13: **if** $\exists t, \text{ s.t. } I_t \neq +\infty$ **then**
14: Find the timeslot t^* with the minimum interference: $t^* = \arg \min_t I_t$.
15: Update the BHTP of satellite m : $\mathbf{x}_m^{t^*} = \mathbf{x}_m^{t^*} \cup \{\tilde{n}_m\}$.
16: **end if**
17: **end for**
18: **until** $\forall m \in \mathcal{M}, \mathbf{X}_{m,temp} = \mathbf{X}_m$.
19: Evenly allocate satellite power over all spotbeams: $p_{mk}^t = P_{sat}/|\mathbf{x}_m^t|, \forall m \in \mathcal{M}, \forall t \in \mathcal{W}$.
Output: The BHTP of each satellite $\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_M$.

B. Joint Multi-Satellite BHTP Design Problem

After solving $\mathcal{P}_2$, we should design the BHTP for each satellite to improve the traffic satisfaction rate of cells. However, obtaining BHTP is a non-convex and NP-hard problem, as shown in $\mathcal{P}_0$. Thus, to further reduce the computational complexity, this letter proposes a low-complexity joint iterative multi-satellite BHTP design method (JIMS-BH). Denote $\mathcal{N}_m$ as the set of cells assigned to satellite m , and the procedure of JIMS-BH is summarized in Alg. 1. The main idea of JIMS-BH is that at each iteration, satellites select the timeslot with minimum total interference to serve the cell with minimum traffic satisfaction rate. Specifically, line 2 to line 4 represent the selection of the cells with the minimum traffic satisfaction rate. Line 5 to line 17 indicate the choice of the least-interfering timeslot. From line 19, the satellite power P_{sat} is distributed evenly over spotbeams. This algorithm terminates when BHTPs are no longer updated, and iterates $K \times W$ times

TABLE I
SIMULATION PARAMETERS

PARAMETERS	VALUES
Longitude range of equatorial region	$[100^\circ E, 105^\circ E]$
Latitude range of equatorial region	$[-1.5^\circ S, 1.5^\circ N]$
NGSO satellite altitude	550 km
Number of NGSO satellite, M	9
Number of GSO satellite	6
GSO satellite location	$80^\circ E : 8^\circ : 120^\circ E$
Number of GSO ground station L	100
Downlink Frequency, f_c	11.7 GHz
Number of cells, N	428
The maximum number of spotbeams, K	8
System bandwidth, B	250 MHz
Total satellite power, P_{sat}	10 dBW
Satellite 3dB beamwidth, θ_b	3°
Maximum transmit antenna gain, G_{max}^{tx}	34.3 dBi
The radius of user antenna's aperture	0.2 m
Maximum receiving antenna gain, G_{max}^{rx}	33.8 dBi
The radius of GSO ground station antenna	0.4 m
The interference threshold, I_{th}	-130 dBW
Noise temperature, T_{noise}	300 K
Number of timeslots in BHTP, W	30

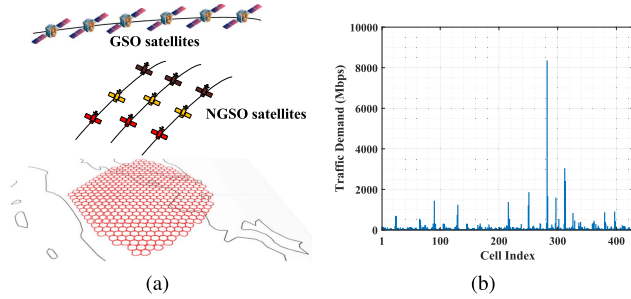


Fig. 3. An illustration of (a) simulation scenario and (b) traffic demands of each cell.

in the worst case. Therefore, JIMS-BH has linear complexity and can maximize the satisfaction of cells as much as possible.

IV. SIMULATION RESULTS AND PERFORMANCE ANALYSIS

A. Simulation Parameter

In this section, we take a specific equatorial region (with a longitude range of $[100^\circ E, 105^\circ E]$ and a latitude range of $[-1.5^\circ S, 1.5^\circ N]$) as an example to evaluate the performance of JIMS-BH. Simulation parameters are detailed in Table I. Some parameter settings refer to Starlink configuration [1]. As shown in Fig. 3(a), at a snapshot, there are 9 satellites distributed on three orbital planes, covering a total of 428 cells. Each satellite can form at most 8 spotbeams. Besides, there are 100 GSO ground stations randomly distributed in this area, which are connected to 6 GSO satellites (located from $80^\circ E$ to $120^\circ E$). Fig. 3(b) shows the aggregate demand of cells, which is obtained by the world population distribution [11]. The beam radiation pattern refers to 3GPP TR 38.811 [12], which has the form as,

$$G(\theta) = \begin{cases} G_{max}, & \theta = 0^\circ \\ G_{max} \left| \frac{4J_1(2\pi a \sin \theta / \lambda)}{2\pi a \sin \theta / \lambda} \right|^2, & 0^\circ < |\theta| \leq 90^\circ \end{cases} \quad (9)$$

where $J_1(\cdot)$ is the Bessel function of the first kind and first order. λ and a represents the wavelength and the radius of the

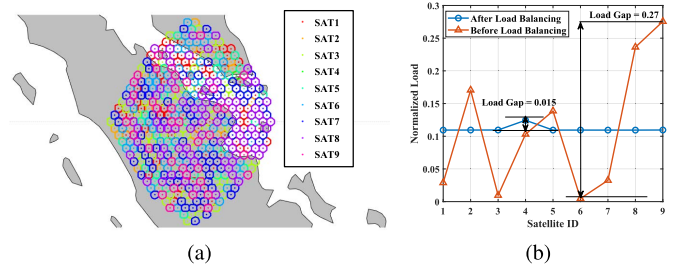


Fig. 4. The results of (a) the satellite-cell assignment and (b) the normalized load of 9 satellites before and after load balancing.

antenna's circular aperture, respectively. θ denotes the off-axis angle in degrees. G_{max} denotes the maximum antenna gain, which has a form as $G_{max} = 10 * \log_{10}[(\frac{2\pi a}{\lambda})^2]$.

In this letter, we give out the upper bound of optimal solution using convex relaxation and compare JIMS-BH with five different BH schemes as follows,

- 1) *The Random Beam Hopping (R-BH)*: In R-BH, each satellite randomly selects 8 cells for service at each timeslot.
- 2) *The Periodic Beam Hopping (P-BH)*: P-BH takes a round-robin scheduling, which means that 9 satellite generates $9 * 8 = 72$ spotbeams and serve 428 cells periodically.
- 3) *Maximum SINR Beam Hopping (MaxSINR-BH)*: In MaxSINR-BH, cells access satellites according to the maximum elevation angle criterion. Then, each satellite applies an approach proposed in [4] to determine BHTP by allocating beams to the timeslot with maximum SINR.
- 4) *Load Balancing and Maximum SINR Beam Hopping (LB-MaxSINR-BH)*: In LB-MaxSINR-BH, load balancing is firstly implemented by solving $\mathcal{P}_2$. Then, each satellite applies an approach proposed in [4] to determine BHTP.
- 5) *Load Balancing and Greedy Beam Hopping (LB-G-BH)*: In LB-G-BH method, load balancing is firstly implemented by solving $\mathcal{P}_2$. Then, each satellite greedily serves the top 8 cells with the most traffic demand at each timeslot.

B. Performance Analysis

1) Performance of Load Balancing:

Fig. 4(a) shows the satellite-cell assignment results by solving $\mathcal{P}_2$. In order to evaluate the load of the satellite, we denote the normalized load as $\hat{S}_m = S_m / \sum_{m=1}^9 S_m$, where S_m is the load of satellite m , which has been defined in (7).

Assuming that before load balancing, cells are assigned to the satellite with the maximum elevation angle, Fig. 4(b) illustrates the normalized load of each satellite before and after load balancing. It can be obtained that the load gap among satellites is reduced from 0.27 to 0.015 after load balancing, indicating that the load variance is reduced by about 80%. Particularly, the load of satellites, which are close to the traffic-intensive area (e.g., satellite 8 and satellite 9), has been effectively lightened by about 54%. Therefore, the proposed satellite-cell assignment can achieve excellent load balancing.

2) Performance of Minimum Traffic Satisfaction Rate:

We denote $\beta_n = \max\{\frac{R_n}{D_n}, 1\}$ as the traffic satisfaction rate of cell n . It is noted that $\beta_n \leq 1$, and the larger the

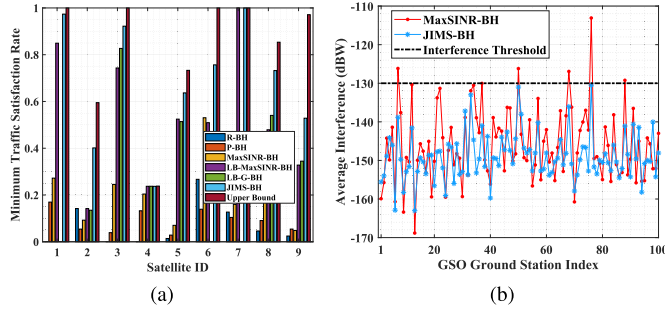


Fig. 5. Simulation results of (a) the minimum traffic satisfaction rate of 9 satellites and (b) the average interference received by 100 GSO ground stations.

TABLE II

UNMET CAPACITY, UNUSED CAPACITY, MISMATCH CAPACITY AND MINIMUM TRAFFIC SATISFACTION RATE OF SIX ALGORITHMS

Algorithm	Total Unmet Capacity (Mbps)	Total Unused Capacity (Mbps)	Total Mismatch Capacity (Mbps)	Minimum Traffic Satisfaction Rate (%)
R-BH	33683	44060	77743	0
P-BH	37829	28364	66193	2.86
MaxSINR-BH	28539	7674	36213	4.85
LB-MaxSINR-BH	14541	8796	23337	14.20
LB-G-BH	16040	11794	27834	0
JIMS-BH	16472	6086	22558	23.72
Upper Bound	—	—	—	23.82

β_n is, the more satisfied the cell is. To better visualization of the simulation results, we further define the minimum traffic satisfaction rate of satellite m as $\hat{\beta}_m = \min_{n \in N_m} \{\beta_n\}$, which represents the worst capacity-demand mismatch among cells of satellite m . Fig. 5(a) shows the minimum traffic satisfaction rate of 9 satellites. It can be seen that JIMS-BH outperforms other benchmarks, where each satellite has the highest traffic satisfaction rate. Besides, in JIMS-BH, the minimum traffic satisfaction rate of satellite 7 equals 1, which means that all cells are satisfied in satellite 7. However, we can observe that in R-BH and LB-G-BH, the minimum traffic satisfaction rate is 0 in some satellites, which indicates that some cells have never been served. Table II summarizes the objective function value (i.e., the minimum traffic satisfaction rate) of different algorithms. Particularly, JIMS-BH has the highest objective function value (i.e., 23.72%). Furthermore, the gap between JIMS-BH and the upper bound of optimal solution is only 0.1%, which illustrates that JIMS-BH can well guarantee of fairness between cells.

3) *Unmet, Unused and Mismatch Capacity*: Different algorithms are evaluated in terms of unmet capacity, unused capacity and mismatch capacity, which are summarized in Table II. The total unmet capacity is denoted by $C_{unmet} = \sum_{n \in N} (D_n - R_n)^+$, where $(x)^+ = \max(0, x)$. The total unused capacity has the form as $C_{unused} = \sum_{n \in N} (R_n - D_n)^+$ and the total mismatch capacity is the sum of unmet and unused capacity. It is noted that JIMS-BH has the lowest unused capacity (i.e., 6086 Mbps), achieving the minimum waste of resources. Although the unmet capacity of JIMS-BH is slightly more than that of LB-G-BH and LB-MaxSINR-BH, JIMS-BH still completes a better offered-requested data match due to the lowest mismatch capacity.

4) *The Interference of GSO Ground Station*: Fig. 5(b) shows the average interference received by 100 GSO ground stations during whole BHTP in terms of JIMS-BH and MaxSINR-BH. We can observe that the interference of JIMS-BH is lower than that of MaxSINR-BH for about 70% ground stations, and the interference is reduced by 5 dBW on average. Since JIMS-BH considers interference avoidance in the cell-satellite assignment phase, the interference is below the interference threshold, i.e., -130 dBW. However, in MaxSINR-BH, cells are assigned to the satellite with the maximum elevation angle, and the connected NGSO satellite may be collinear with GSO satellites, resulting in nonnegligible interference.

V. CONCLUSION

This letter has studied a multi-satellite communication system, where a beam hopping strategy aiming to match uneven traffic demands and avoid interference to GSO ground stations is proposed. In order to solve the complex optimization problem, we decompose it into two sub-problems. Firstly, load balancing and interference avoidance are achieved via rationally assigning connections between satellites and cells. Then, an iterative beam allocation scheme based on minimizing inter-beam interference is adopted to meet differentiated demands. Simulation results have verified the superiority of the proposed method in guaranteeing the service fairness of cells and reducing the interference to ground stations.

REFERENCES

- [1] I. D. Portillo, B. G. Cameron, and E. F. Crawley, "A technical comparison of three low earth orbit satellite constellation systems to provide global broadband," *Acta Astronaut.*, vol. 159, pp. 123–135, Jun. 2019.
- [2] J. Du, C. Jiang, H. Zhang, X. Wang, Y. Ren, and M. Debbah, "Secure satellite-terrestrial transmission over incumbent terrestrial networks via cooperative beamforming," *IEEE J. Sel. Areas Commun.*, vol. 36, no. 7, pp. 1367–1382, Jul. 2018.
- [3] *Digital Video Broadcasting (DVB); Second Generation Framing Structure, Channel Coding and Modulation Systems for Broadcasting, Interactive Services, News Gathering and Other Broadband Satellite Applications; Part 2: DVB-S2 Extensions (DVB-S2X)*, Standard ETSI TR 102 376-2, Jan. 2021.
- [4] R. Alegre-Godoy, N. Alagha, and M. A. Vazquez-Castro, "Offered capacity optimization mechanisms for multi-beam satellite systems," in *Proc. IEEE Int. Conf. Commun. (ICC)*, Jun. 2012, pp. 3180–3184.
- [5] B. Deng, C. Jiang, J. Wang, and L. Kuang, "Beam scheduling with various mission demands in data relay satellite systems," *J. Commun. Inf. Netw.*, vol. 6, no. 4, pp. 396–410, Dec. 2021.
- [6] L. Lei, E. Lagunas, Y. Yuan, M. G. Kibria, S. Chatzinotas, and B. Ottersten, "Beam illumination pattern design in satellite networks: Learning and optimization for efficient beam hopping," *IEEE Access*, vol. 8, pp. 136655–136667, 2020.
- [7] Z. Lin, Z. Ni, L. Kuang, C. Jiang, and Z. Huang, "Dynamic beam pattern and bandwidth allocation based on multi-agent deep reinforcement learning for beam hopping satellite systems," *IEEE Trans. Veh. Technol.*, vol. 71, no. 4, pp. 3917–3930, Apr. 2022.
- [8] J. Zhang, D. Qin, C. Kong, F. Zhao, R. Li, and J. Wang, "System-level evaluation of beam hopping in NR-based LEO satellite communication system," 2022, *arXiv:2205.10721*.
- [9] X. Hu, L. Wang, Y. Wang, S. Xu, Z. Liu, and W. Wang, "Dynamic beam hopping for DVB-S2X GEO satellite: A DRL-powered GA approach," *IEEE Commun. Lett.*, vol. 26, no. 4, pp. 808–812, Apr. 2022.
- [10] M. Grant and S. Boyd. (Sep. 2013). *CVX: MATLAB Software for Disciplined Convex Programming, Version 2.0 Beta*. [Online]. Available: <http://cvxr.com/cvx>
- [11] *Global High Resolution Population Denominators Project*, WorldPop, Center Int. Earth Sci. Inf. Netw., New York, NY, USA, 2018.
- [12] *Study on New Radio (NR) to Support Non-Terrestrial Networks*, document TR 38.811, 3GPP, 2019.